

The Circle

Definition

A circle is the locus of a point in a plane which moves so that its distance from a given point is always equal to a given distance. The given point is called the centre and the given distance is the radius of the circle.

Art. 1 The equation of a circle with centre at (h, k) and radius a is

$$(x - h)^2 + (y - k)^2 = a^2 \quad (1)$$

Corollary: If the centre of the circle is the origin, then (1) becomes

$$x^2 + y^2 = a^2 \quad (2)$$

Art. 2 Equation of the circle in polar coordinates

Let the centre C of the circle be at $C(r_1, \theta_1)$ and a be the radius of the circle. Let $P(r, \theta)$ be any point on the circle. Then $\angle COP = \theta - \theta_1$ and from $\triangle COP$, we have

$$CP^2 = OC^2 + OP^2 - 2(OC)(OP)\cos COP.$$

Thus the equation of the circle in terms of polar coordinates is

$$r_1^2 + r^2 - 2r_1r \cos(\theta - \theta_1) = a^2 \quad (3)$$

Corollary 1: If the pole lies on the circle, then $r_1 = a$. In this case, (3) becomes

$$r = 2a \cos(\theta - \theta_1) \quad (4)$$

Corollary 2: If $\theta_1 = 0^\circ$, we see that the initial line is a diameter. In this case, the equation of the circle with radius a is

$$r = 2a \cos \theta \quad (5)$$

Corollary 3: If the centre is the pole, then the equation of the circle is $r = a$.

Art. 3 General equation of a circle

The general equation of a circle with centre at $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$ is

$$x^2 + y^2 + 2gx + 2fy + c = 0. \quad (6)$$

Art. 4 The equation of the circle whose diameter is the line joining the points (x_1, y_1) and (x_2, y_2) is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0. \quad (7)$$

Art. 5 Position of a point with respect to a circle

Any point (x_1, y_1) will be outside, on or inside the circle (6) according as its distance from the centre $(-g, -f)$ is greater, equal, or less than the radius $\sqrt{g^2 + f^2 - c}$.

Art. 6 The equation of the tangent to the circle (6) at (x_1, y_1) is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0. \quad (8)$$

Corollary: The equation of the tangent to the circle (2) at (x_1, y_1) is

$$xx_1 + yy_1 = a^2. \quad (9)$$

Art. 7 The equation of the normal to the circle (6) at (x_1, y_1) is

$$y(x_1 + g) - x(y_1 + f) + fx_1 - gy_1 = 0. \quad (10)$$

Corollary: The equation of the normal to the circle (2) at (x_1, y_1) is

$$xy_1 - yx_1 = 0. \quad (11)$$

Art. 8 Length and mid-point of the chord intercepted by the circle

$$x^2 + y^2 = a^2 \quad (1)$$

on the line

$$y = mx + c. \quad (2)$$

Let the equation of the chord AB be $y = mx + c$ and let C be the mid-point of the chord.

Therefore, the equation of OC is

$$my + x = 0. \quad (3)$$

Solving (2) and (3), we get $C\left(\frac{-mc}{1+m^2}, \frac{c}{1+m^2}\right)$.

From (1) and (2), we get

$$\begin{aligned} x^2 + (mx + c)^2 &= a^2 \\ \Rightarrow (1 + m^2)x^2 + 2mcx + c^2 - a^2 &= 0 \end{aligned} \quad (4)$$

Let x_1, x_2 be the roots of (4). Then $x_1 + x_2 = -\frac{2mc}{1+m^2}$, and $x_1x_2 = \frac{c^2 - a^2}{1+m^2}$.

$$\begin{aligned} \text{Hence } x_1 - x_2 &= \sqrt{(x_1 + x_2)^2 - 4x_1x_2} = \frac{2}{1+m^2} \sqrt{m^2c^2 - (c^2 - a^2)(1+m^2)} \\ &= \frac{2}{1+m^2} \sqrt{a^2(1+m^2) - c^2}. \end{aligned}$$

Let y_1, y_2 be the ordinates of A and B , respectively. Since (x_1, y_1) and (x_2, y_2) are on (2), we have $y_1 - y_2 = (mx_1 + c) - (mx_2 + c) = m(x_1 - x_2)$.

$$\text{Hence } AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{1+m^2} (x_1 - x_2) = 2\sqrt{\frac{a^2(1+m^2) - c^2}{1+m^2}}.$$

Note: The general equation of second degree in x and y

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

will represent a circle if $a = b$ and $h = 0$.

Art. 9 Chord whose mid-point is given

To find the equation of the chord of a circle whose middle point is given.

Let the equation of the circle be $x^2 + y^2 = a^2$ (1)

Let (x_1, y_1) be the mid-point of any chord of (1).

The equation of any chord passing through (x_1, y_1) is $y - y_1 = m(x - x_1)$ (2)

Also, the equation of the line joining (x_1, y_1) and centre of (1) is $y = \frac{y_1}{x_1}x$ (3)

If (2) and (3) are perpendicular, then $m \frac{y_1}{x_1} = -1 \Rightarrow m = -\frac{x_1}{y_1}$

$$\begin{aligned}
\text{So (2)} \Rightarrow y - y_1 &= -\frac{x_1}{y_1}(x - x_1) \\
\Rightarrow yy_1 - y_1^2 &= -xx_1 + x_1^2 \\
\Rightarrow xx_1 + yy_1 &= x_1^2 + y_1^2 \quad (4)
\end{aligned}$$

which is the required equation of the chord.

$$\begin{aligned}
\text{Note: (4) can be written symbolically as } T &= S_1 \quad (5) \\
\text{where } T &= xx_1 + yy_1 - a^2 \text{ and } S_1 = x_1^2 + y_1^2 - a^2.
\end{aligned}$$

Art. 10 To find the length of either of the tangents from an external point to a circle.

Let the external point be $P(x_1, y_1)$ and the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$. Therefore, the centre of the circle is $C(-g, -f)$ and radius, $CT = \sqrt{g^2 + f^2 - c}$ where T is the point of contact to the circle. From the right angled triangle ΔCTP , we have

$$\begin{aligned}
CP^2 &= PT^2 + CT^2 \\
\Rightarrow PT^2 &= CP^2 - CT^2 = [(x_1 + g)^2 + (y_1 + f)^2] - (g^2 + f^2 - c) \\
&= x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c.
\end{aligned}$$

Corollary: The length of the tangent T from an external point (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is given by $T^2 = x_1^2 + y_1^2 - a^2$.

Art. 11 Condition of tangency of a line

Find the condition that the line

$$y = mx + c \quad (1)$$

is a tangent to the circle

$$x^2 + y^2 + 2gx + 2fy + c' = 0 \quad (2)$$

and the circle

$$x^2 + y^2 = a^2 \quad (3)$$

From (1) and (2), we have

$$\begin{aligned}
x^2 + (mx + c)^2 + 2gx + 2f(mx + c) + c' &= 0 \\
\Rightarrow (1 + m^2)x^2 + 2(mc + mf + g)x + (2fc + c^2 + c') &= 0 \quad (4)
\end{aligned}$$

which is quadratic in x .

Now if the line (1) is a tangent to the circle (2), the line will cut the circle at two coincident points. In this case, the roots of (4) will be equal.

$$\begin{aligned}
\text{Thus, } 4(mc + mf + g)^2 &= 4(1 + m^2)(2fc + c^2 + c') \\
\Rightarrow (mc + mf + g)^2 &= (1 + m^2)(2fc + c^2 + c') \quad (5)
\end{aligned}$$

which is the required condition.

For the circle $x^2 + y^2 = a^2$, we have $f = g = 0$ and $c_1 = -a^2$. Then (5) becomes

$$\begin{aligned}
m^2 c^2 &= (1 + m^2)(c^2 - a^2) \Rightarrow c^2 = a^2(1 + m^2) \\
\Rightarrow c &= \pm a\sqrt{1 + m^2} \quad (6)
\end{aligned}$$

Thus, (1) is a tangent to the circle (3) if $c = a\sqrt{1+m^2}$ or $c = -a\sqrt{1+m^2}$.

In other words, the lines

$$y = mx \pm a\sqrt{1+m^2} \quad (7)$$

are always tangents to the circle $x^2 + y^2 = a^2$. It follows that corresponding to any value of m , we can find two tangents to the circle $x^2 + y^2 = a^2$.

Art. 12 Chord of contact

If two tangents be drawn to a circle from an external point, the straight line joining the points of contact is called the chord of contact of the tangents drawn from the point.

In the adjacent figure, the tangents PT and PT' are drawn from P where T, T' are the points of contact to the circle. So the line TT' is the chord of contact to the circle.

Art. 13 Equations of chord of contact

The equation of the chord of contact of the tangents drawn from (x_1, y_1) to the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ is } xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0. \quad (8)$$

Corollary: Let the equation of the circle be $x^2 + y^2 = a^2$ and coordinates of the external point P be (x_1, y_1) . Let the coordinates of the point of contact of tangents drawn from (x_1, y_1) be (x_2, y_2) and (x_3, y_3) .

The equation of the tangent to $x^2 + y^2 = a^2$ at (x_2, y_2) is $xx_2 + yy_2 = a^2$ and that at (x_3, y_3) is $xx_3 + yy_3 = a^2$.

Pole and Polar

Art. 14 Equation of the polar of a point

The equation of the polar of the point (x_1, y_1) with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is given by the equation

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

Note: The equation of the polar of the point (x_1, y_1) with respect to the circle $x^2 + y^2 = a^2$ is given by the equation

$$xx_1 + yy_1 = a^2.$$

Art. 15 Pole of a line

The pole of a line $lx + my + n = 0$ with respect to the circle $x^2 + y^2 = a^2$ is $\left(\frac{-a^2l}{n}, \frac{-a^2m}{n}\right)$ and

the pole of a line $lx + my + n = 0$ with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\left(-g + \frac{l(g^2 + f^2 - c)}{lg + mf - n}, -f + \frac{m(g^2 + f^2 - c)}{lg + mf - n}\right)$.

Art. 16 Conjugate points and conjugate lines

If two points are such that the polar of one passes through the other, then the points are called *conjugate points*. If two lines are such that the pole of one lies on the other, then such lines are called *conjugate lines*.

Art. 17 Combined equation of pair of tangents**Example 1**

Find the pole of the straight line $2x - y = 6$ with respect to the circle $5x^2 + 5y^2 = 9$.

Solution. Let $P(x_1, y_1)$ be the pole. Then the equation of the polar with respect to the circle $5x^2 + 5y^2 = 9$ is $5xx_1 + 5yy_1 = 9$ (1)

Since (1) is identical with the given line, we have

$$\frac{5x_1}{2} = \frac{5y_1}{-2} = \frac{9}{6} \Rightarrow x_1 = \frac{3}{5} \text{ and } y_1 = -\frac{3}{10}.$$

The required pole is $\left(\frac{3}{5}, -\frac{3}{10}\right)$.

Example 2

Show that the equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $c(x^2 + y^2) = (gx + fy)^2$.

Solution. The equation to the pair of tangents is $SS_1 = T^2$

$$\begin{aligned} (x^2 + y^2 + 2gx + 2fy + c)c &= [x.0 + y.0 + g(x+0) + f(y+0) + c]^2 \\ \Rightarrow c(x^2 + y^2) + c(2gx + 2fy + c) &= (gx + fy + c)^2 \\ &= (gx + fy)^2 + 2(gx + fy)c + c^2 \end{aligned}$$

$$\Rightarrow c(x^2 + y^2) = (gx + fy)^2.$$

Example 3

Prove that if the polar of a point P with respect to the circle $x^2 + y^2 = 37$ touches the circle $(x-3)^2 + (y+2)^2 = 25$ then the locus of P must be a conic.

Solution. Given circles are $x^2 + y^2 = 37$ (1)

$$\text{and } (x-3)^2 + (y+2)^2 = 25 \quad (2)$$

Let $P(x_1, y_1)$ be the pole.

Then the polar of P with respect to the circle $x^2 + y^2 = 37$ is $xx_1 + yy_1 = 37$ (3)

Since (1) touches (2), we have

$$\frac{3x_1 - 2y_1 - 37}{\sqrt{x_1^2 + y_1^2}} = 5$$

(length of the perpendicular from (3, -2) to (1) is equal to the radius of the circle)

So that $(3x_1 - 2y_1 - 37)^2 = 25(x_1^2 + y_1^2)$

$$\Rightarrow 16x_1^2 + 21y_1^2 + 12x_1y_1 + 222x_1 - 148y_1 - 1369 = 0$$

Therefore, the required locus is

$$16x + 21 + 12xy + 222x - 148y - 1369 = 0.$$

Example 4

Find the equation of the direct common tangents to the circles $x^2 + y^2 = 16$ and $x^2 + y^2 + 6x - 8y = 0$.

Solution. The centre of the two circles are (0,0) and (-3,4) and their radii are 4 and 5, respectively. Let (x_1, y_1) be the point which divide the line of centers externally in the ratio 4:5. Therefore, we have

$$x_1 = \frac{5 \times 0 - 4(-3)}{5 - 4} = 12, \quad y_1 = \frac{5 \times 0 - 4 \times 4}{5 - 4} = -16.$$

So the direct tangents pass through (12, -16). Hence their equations are given by

$$y + 16 = m(x - 12)$$

These lines may touch the circle if the perpendicular from the centre of the circle is equal to the radius 4.

$$\text{Thus } \frac{12m + 16}{\sqrt{1 + m^2}} = \pm 4 \Rightarrow 8m^2 + 24m + 15 = 0 \Rightarrow m = \frac{-24 \pm \sqrt{576 - 480}}{16} = \frac{-6 \pm \sqrt{6}}{4}$$

Hence the required equations are

$$\begin{aligned} y + 16 &= \frac{-6 \pm \sqrt{6}}{4} (x - 12) \\ \Rightarrow 4y + 64 &= -6(x - 12) \pm \sqrt{6}(x - 12) \\ \Rightarrow 6x + 4y - 8 &= \pm \sqrt{6}(x - 12). \end{aligned}$$

Example 5

Find the equation of the transverse common tangents to the circles $x^2 + y^2 - x - 2y + 1 = 0$ and $x^2 + y^2 - 3x + 4y = 0$.

Solution. The centre of the circles are $(\frac{1}{2}, 1)$ and $(\frac{3}{4}, -1)$ and their radii are $\frac{1}{2}$ and $\frac{5}{4}$, respectively. Let (x_1, y_1) be the point which divide the line of centers internally in the ratio $\frac{1}{2} : \frac{5}{4}$.

Therefore, we have

$$x_1 = \frac{\frac{5}{4} \times \frac{1}{2} + \frac{1}{2} \times \frac{3}{4}}{\frac{5}{4} + \frac{1}{2}} = \frac{4}{7}, \quad y_1 = \frac{\frac{5}{4} \times 1 + \frac{1}{2}(-1)}{\frac{5}{4} + \frac{1}{2}} = \frac{3}{7}$$

The equation of transverse tangents are $y - \frac{3}{7} = m(x - \frac{4}{7})$

$$\text{We can write } \frac{1 - \frac{3}{7} - m(\frac{1}{2} - \frac{4}{7})}{\sqrt{1 + m^2}} = \pm \frac{1}{2}$$

$$\begin{aligned} \Rightarrow \left(\frac{4}{7} + \frac{m}{14} \right)^2 &= \frac{1}{4}(1 + m^2) \Rightarrow (8 + m)^2 = \frac{196}{4}(1 + m^2) \\ \Rightarrow 48m^2 - 16m - 15 &= 0 \end{aligned}$$

$$\Rightarrow m = \frac{3}{4}, -\frac{5}{12}.$$

Hence the required equations are

$$y - \frac{3}{7} = \frac{3}{4}\left(x - \frac{4}{7}\right) \quad \text{and} \quad y - \frac{3}{7} = -\frac{5}{12}\left(x - \frac{4}{7}\right)$$

$$\Rightarrow 4y = 3x \quad \Rightarrow 5x + 12y = 8.$$

Example 6

Find the locus of the middle points of the chords of $x^2 + y^2 = a^2$ which subtend a right angle at the centre.

Solution. Let (h, k) be the middle-point of any one of the chords which subtend a right angle at the centre of $x^2 + y^2 = a^2$. The equation of the chord in terms of the middle-point (h, k) is $T = S_1$

$$\Rightarrow hx + ky - a^2 = h^2 + k^2 - a^2 \Rightarrow \frac{hx + ky}{h^2 + k^2} = 1 \quad (1)$$

$$\text{The center of the circle} \quad x^2 + y^2 = a^2 \quad (2)$$

is $(0, 0)$. So the equation of the pair of straight lines joining the origin to the point of intersection of (1) and (2) is

$$x^2 + y^2 - a^2 \left(\frac{hx + ky}{h^2 + k^2} \right)^2 = 0$$

[making (2) homogeneous with the help of (1)]

These lines will subtend a right angle at the centre if

$$\begin{aligned} & \text{coefficient of } x^2 + \text{coefficient of } y^2 = 0 \\ \Rightarrow & \{(h^2 + k^2)^2 - a^2 h^2\} + \{(h^2 + k^2)^2 - a^2 k^2\} = 0 \\ \Rightarrow & 2(h^2 + k^2)^2 - a^2(h^2 + k^2) = 0 \\ \Rightarrow & (h^2 + k^2)\{2(h^2 + k^2) - a^2\} = 0 \\ \Rightarrow & h^2 + k^2 - a^2 = 0 \text{ since } h^2 + k^2 \neq 0 \end{aligned}$$

Putting $h = x, k = y$, we get $x^2 + y^2 = a^2$ which is the required locus.

Example 7

From any point on one given circle, tangents are drawn to another given circle. Prove that the locus of the middle-point of chord of contact is a third circle.

Solution. Let the equation of the given circles be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (1)$$

$$x^2 + y^2 = a^2 \quad (2)$$

$$\text{Let } P(x_1, y_1) \text{ be any point on (1). Then } x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0 \quad (3)$$

Let QR be the chord of contact where Q and R are on (2). So the equation of QR is

$$xx_1 + yy_1 - a^2 = 0 \quad (4)$$

If $M(h, k)$ be the mid-point of QR , then

$$\begin{aligned} T = S_1 & \Rightarrow hx + ky - a^2 = h^2 + k^2 - a^2 \\ \Rightarrow hx + ky - (h^2 + k^2) & = 0 \end{aligned} \quad (5)$$

Since (4) and (5) represent the same line, we have

$$\begin{aligned} \frac{x_1}{h} = \frac{y_1}{k} & = \frac{a^2}{h^2 + k^2} \\ \Rightarrow x_1 & = \frac{a^2 h}{h^2 + k^2}, y_1 = \frac{a^2 k}{h^2 + k^2} \end{aligned}$$

Then (1) can be written in the form as

$$\left(\frac{a^2 h}{h^2 + k^2}\right)^2 + \left(\frac{a^2 k}{h^2 + k^2}\right)^2 + 2g \frac{a^2 h}{h^2 + k^2} + 2f \frac{a^2 k}{h^2 + k^2} + c = 0$$

$$\Rightarrow c(h^2 + k^2) + 2ga^2 h + 2fa^2 k + a^4 = 0$$

Thus, the locus of (h,k) is

$$c(x^2 + y^2) + 2ga^2 x + 2fa^2 y + a^4 = 0.$$

Exercises

- Find the equation of the circle passing through $(5,7)$ and $(6,6)$ and has its centre on $2x-y+1=0$.
- Find the equations of the tangents to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the straight line $4x + 3y + 5 = 0$.
- Show that the straight line $3x + 4y + 20 = 0$ touches the circle $x^2 + y^2 = 16$ and find the point of contact.
- The equation of a chord of the circle $x^2 + y^2 + 4x - 6y = 0$ is $x + 2y = 0$. Find the equation of the circle described on this chord as diameter.
- Find the length of the chord intercepted by the circle $x^2 + y^2 = a^2$ on the line $y = mx + c$.
- Find the equation of the circle which passes through the origin and makes intercepts a and b on the coordinate axes.
- $ABCD$ is a square with side of length ' l '. If AB and AD be taken as coordinate axes, show that the circle circumscribing the square will be $x^2 + y^2 = l(x + y)$.
- Show that the locus of the foot of the perpendicular from the origin to the line which always passes through a fixed point (h,k) is a circle.
- Find the direct and transverse common tangents to the circle $x^2 + y^2 - 22x + 4y + 100 = 0$ and $x^2 + y^2 + 22x - 4y - 100 = 0$.
- Find the equation of the transverse common tangents to the circles $x^2 + y^2 - 4x - 2y + 4 = 0$ and $x^2 + y^2 + 4x + 2y - 4 = 0$.
- The polar of the point P with respect to the circle $x^2 + y^2 - 2bx = 0$ touches the circle $x^2 + y^2 = a^2$. Find the locus of P .
- Find the pole of the line $x + 2y + 1 = 0$ with respect to the circle $x^2 + y^2 - 4x + 6y - 11 = 0$.